

OverTheWire – Krypton

Proof of Concept Report

By

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This report contains all levels of the **Krypton wargame** on OverTheWire.

Krypton focuses on **basic cryptography concepts** such as classical ciphers and encryption weaknesses.

Link to Krypton wargame:

<https://overthewire.org/wargames/krypton/>

Krypton Level 0 → Level 1

The following string encodes the password using Base64:

“S1JZUFRPTkITR1JFQVQ=”, Use this password to log in to
krypton.labs.overthewire.org with username krypton1 using SSH on port 2231.

Command to encode :echo

Steps:

```
echo "S1JZUFRPTkITR1JFQVQ=" | base64 -d
```

```
(shrutirane@shrutirane)-[~]  
$ echo "S1JZUFRPTk1TR1JFQVQ=" | base64 -d  
KRYPTONISGREAT
```

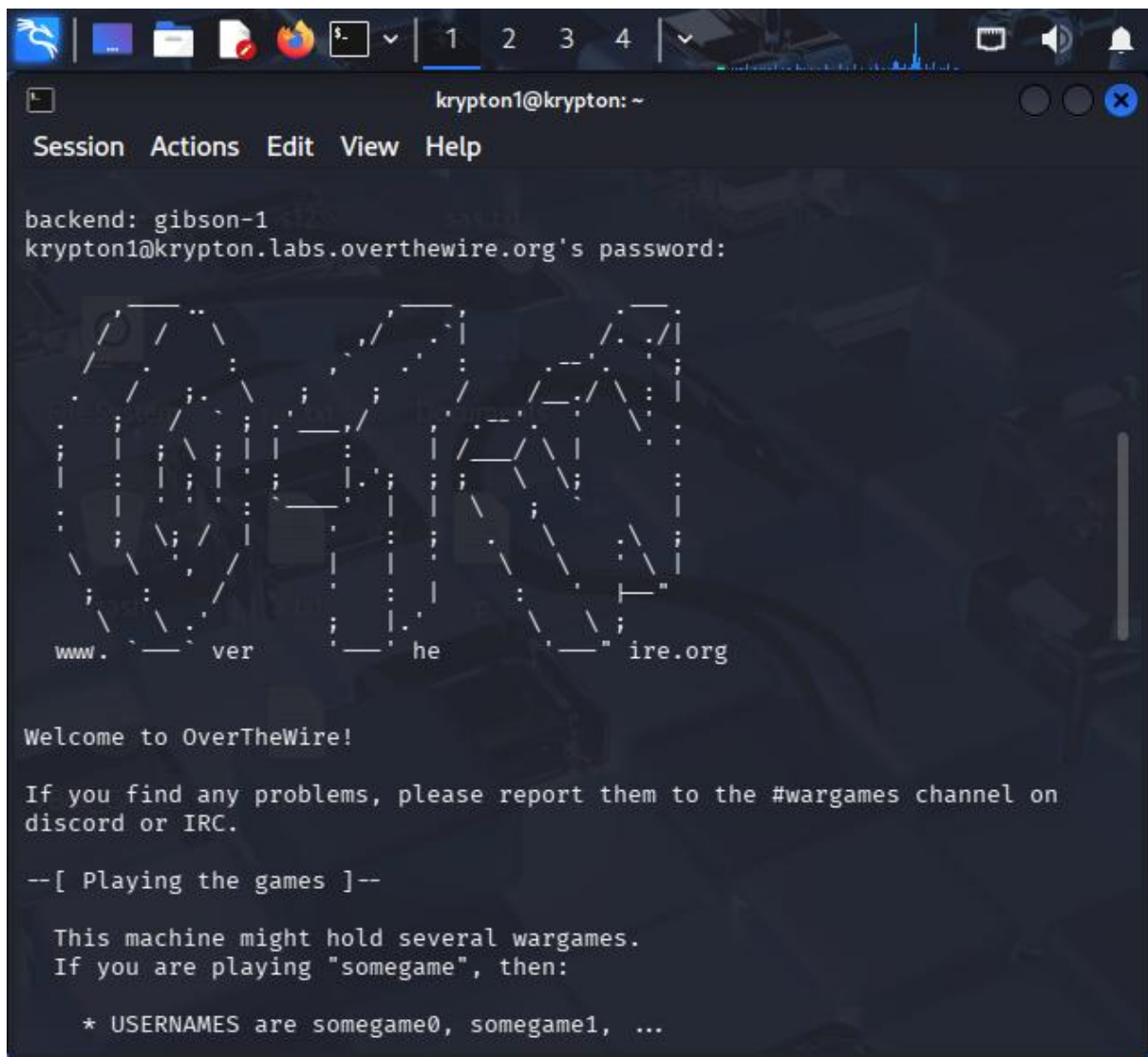
```
(shrutirane@shrutirane)-[~]  
$ ssh krypton1@krypton.labs.overthewire.org -p 2231
```

Konfigurationsmatrix

This is an OverTheWire game server.
More information on <http://www.overthewire.org/wargames>

```
backend: gibbon-1
krypton1@krypton.labs.overthewire.org's password:
```

```
(shrutirane@shrutirane)-[~]  
$ ssh krypton1@krypton.labs.overthewire.org -p 2231
```

A screenshot of a terminal window titled 'krypton1@krypton: ~'. The window has a menu bar with 'Session', 'Actions', 'Edit', 'View', and 'Help'. The terminal output shows the user 'gibson-1' logging in to 'krypton1@krypton.labs.overthewire.org'. A large ASCII art logo for 'OverTheWire' is displayed, with the URL 'www.overthewire.org' below it. The terminal then displays a welcome message and instructions for finding wargames, including a list of usernames: 'somegame0, somegame1, ...'.

```
krypton1@krypton: ~
Session Actions Edit View Help

backend: gibson-1
krypton1@krypton.labs.overthewire.org's password:

Welcome to OverTheWire!

If you find any problems, please report them to the #wargames channel on
discord or IRC.

--[ Playing the games ]--

This machine might hold several wargames.
If you are playing "somegame", then:

* USERNAMES are somegame0, somegame1, ...
```

Krypton Level 1 → Level 2

The password for level 2 is in the file 'krypton2'. It is 'encrypted' using a simple rotation. It is also in non-standard ciphertext format. When using alpha characters for cipher text .

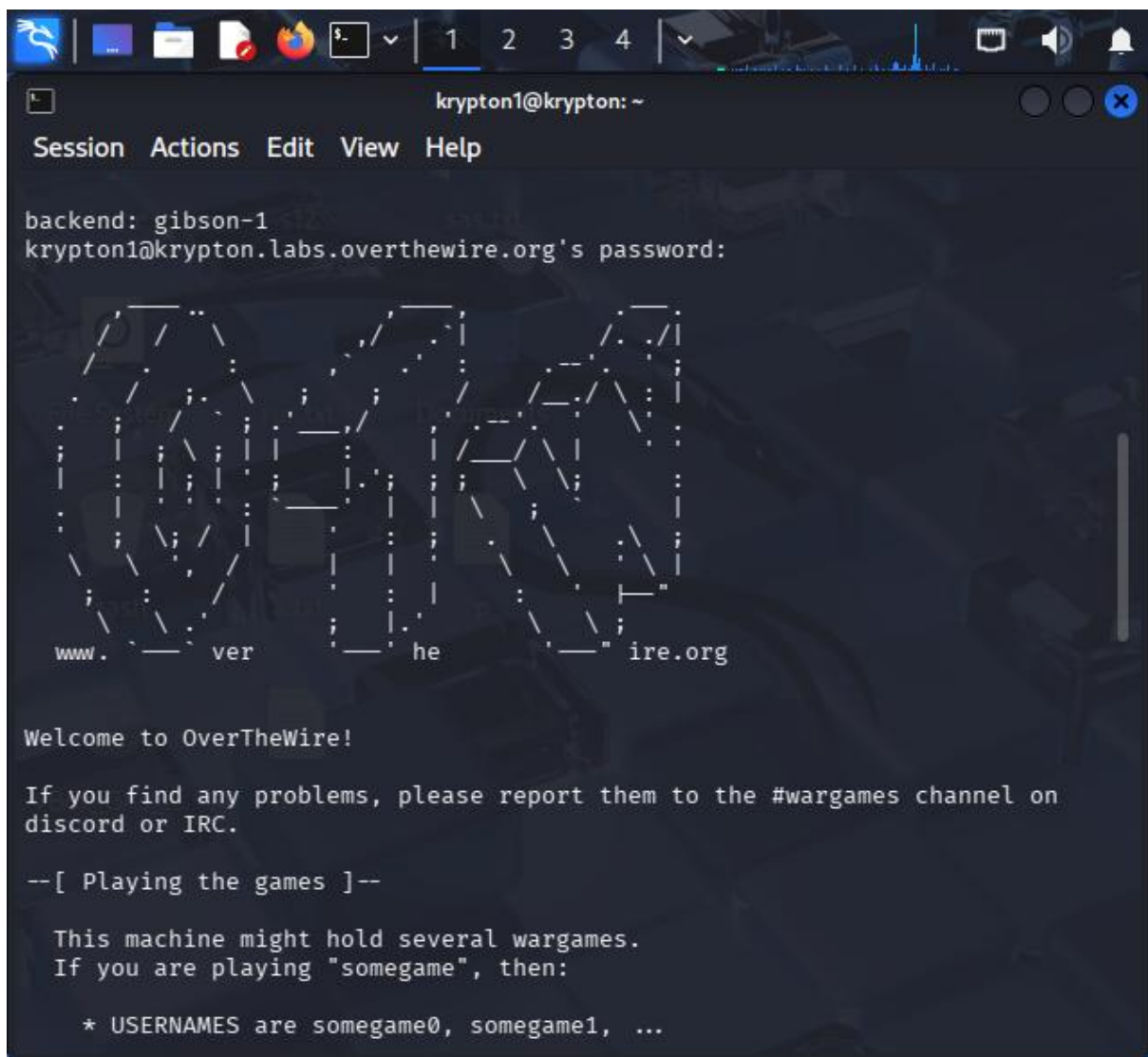
command: ls-la,cd,ls,cat

Steps:

Login to the Krypton server using SSH.

List files in the home directory.

1. Move to the root directory to locate Krypton folders.
2. Navigate to the Krypton directory.
3. Enter the directory for level 1.
4. Read the README file to understand the task.
5. Read the encrypted password file for the next level.
6. The output is ROT13 encrypted text, Decode the ROT13 text using the cryptii tool.
7. Logout from the current session.
8. Login to the next level using the decoded password.



The screenshot shows a terminal window titled 'krypton1@krypton: ~'. The window has a menu bar with 'Session', 'Actions', 'Edit', 'View', and 'Help'. The terminal output is as follows:

```
backend: gibson-1
krypton1@krypton.labs.overthewire.org's password:
www. ver he ire.org

Welcome to OverTheWire!

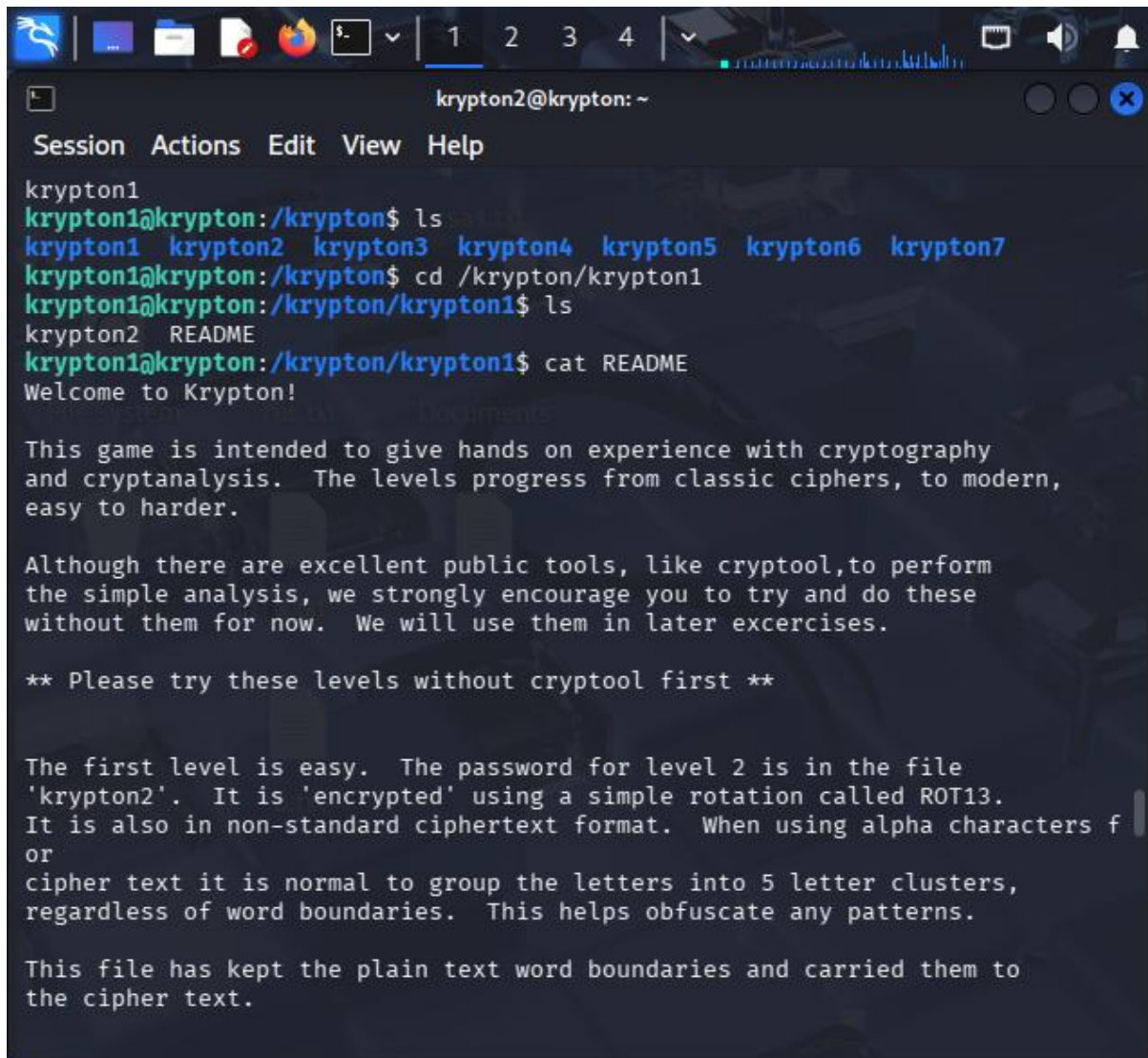
If you find any problems, please report them to the #wargames channel on
discord or IRC.

--[ Playing the games ]--

This machine might hold several wargames.
If you are playing "somegame", then:

* USERNAMES are somegame0, somegame1, ...
```

```
krypton2@krypton: ~  
Session Actions Edit View Help  
krypton1@krypton:~$ ls  
krypton1@krypton:~$ ls -la  
total 20  
drwxr-xr-x  2 root root 4096 Oct 14 09:27 .  
drwxr-xr-x 150 root root 4096 Oct 14 09:29 ..  
-rw-r--r--  1 root root  220 Mar 31  2024 .bash_logout  
-rw-r--r--  1 root root 3851 Oct 14 09:19 .bashrc  
-rw-r--r--  1 root root  807 Mar 31  2024 .profile  
krypton1@krypton:~$ cd /..  
krypton1@krypton:/$ ls  
behemoth      krypton        media          snap  
bin            lib            mnt            srv  
bin.usr-is-merged lib32          narnia         sys  
boot          lib64          opt            tmp  
dev           lib.usr-is-merged proc           usr  
drifter       libx32         root           utumno  
etc           lost+found     run            var  
formulaone    manpage       sbin           vortex  
home          maze          sbin.usr-is-merged  
krypton1@krypton:/$ cd /krypton  
krypton1@krypton:/krypton$ ls  
krypton1  krypton2  krypton3  krypton4  krypton5  krypton6  krypton7  
krypton1@krypton:/krypton$ cat krypton2  
cat: krypton2: Is a directory  
krypton1@krypton:/krypton$ cat /krypton2  
cat: /krypton2: No such file or directory  
krypton1@krypton:/krypton$ cd /krypton1  
-bash: cd: /krypton1: No such file or directory  
krypton1@krypton:/krypton$ whoami  
krypton1
```

A terminal window titled 'krypton2@krypton: ~' with a menu bar (Session, Actions, Edit, View, Help) and a dark background. The terminal shows a user named 'krypton1' navigating through a directory structure. The user lists files in the current directory, changes to the 'krypton' subdirectory, and then lists files there, finding a 'README' file. The user then displays the contents of the README file, which provides instructions for the Krypton game, including a warning to try levels without 'cryptool' first and details about the first level's encryption (ROT13) and formatting (5-letter clusters).

```
krypton1
krypton1@krypton:/krypton$ ls
krypton1  krypton2  krypton3  krypton4  krypton5  krypton6  krypton7
krypton1@krypton:/krypton$ cd /krypton/krypton1
krypton1@krypton:/krypton/krypton1$ ls
krypton2  README
krypton1@krypton:/krypton/krypton1$ cat README
Welcome to Krypton!

This game is intended to give hands on experience with cryptography
and cryptanalysis. The levels progress from classic ciphers, to modern,
easy to harder.

Although there are excellent public tools, like cryptool, to perform
the simple analysis, we strongly encourage you to try and do these
without them for now. We will use them in later excercises.

** Please try these levels without cryptool first **

The first level is easy. The password for level 2 is in the file
'krypton2'. It is 'encrypted' using a simple rotation called ROT13.
It is also in non-standard ciphertext format. When using alpha characters f
or
cipher text it is normal to group the letters into 5 letter clusters,
regardless of word boundaries. This helps obfuscate any patterns.

This file has kept the plain text word boundaries and carried them to
the cipher text.
```

```
krypton2@krypton: ~  
Session Actions Edit View Help  
** Please try these levels without cryptool first **  
  
The first level is easy. The password for level 2 is in the file  
'krypton2'. It is 'encrypted' using a simple rotation called ROT13.  
It is also in non-standard ciphertext format. When using alpha characters f  
or  
cipher text it is normal to group the letters into 5 letter clusters,  
regardless of word boundaries. This helps obfuscate any patterns.  
  
This file has kept the plain text word boundaries and carried them to  
the cipher text.  
  
Enjoy!  
krypton1@krypton:/krypton/krypton1$ cat krypton2  
YRIRY GJB CNFFJBEQ EBGGRA  
krypton1@krypton:/krypton/krypton1$ exit  
logout  
Connection to krypton.labs.overthewire.org closed.
```

https://v2.cryptii.com/rot13/text

INTERPRET AS
ROT13 ▼

CONVERT TO
TEXT ▼

Variant

ROT13 (letters)

YRIRY GJB CNFFJBEQ EBGGRA

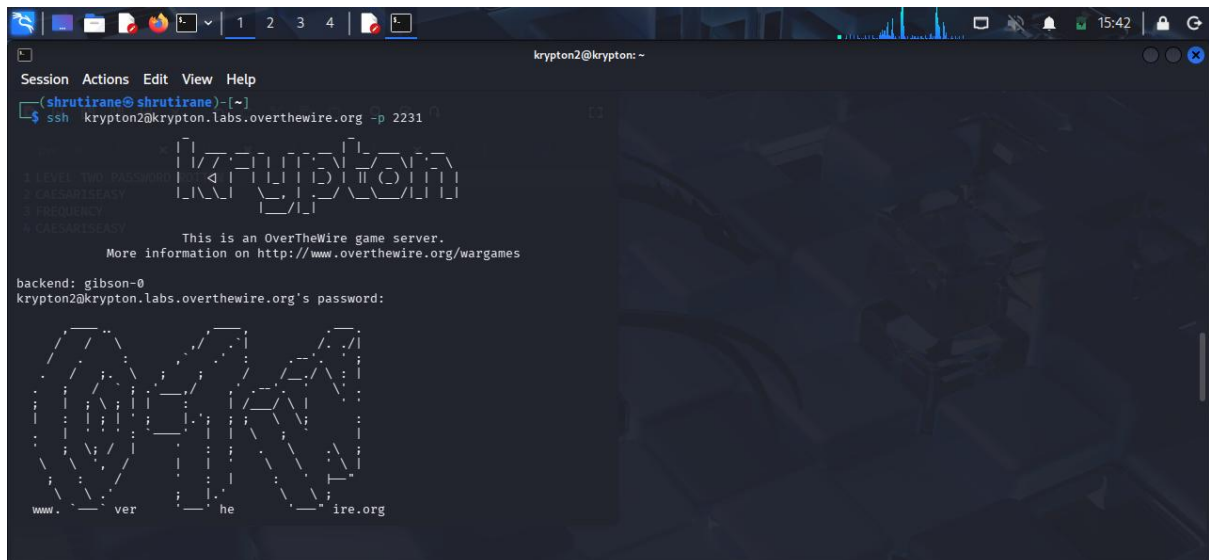
<> | 25

Transform

None

LEVEL TWO PASSWORD ROTTEN

<> | 25



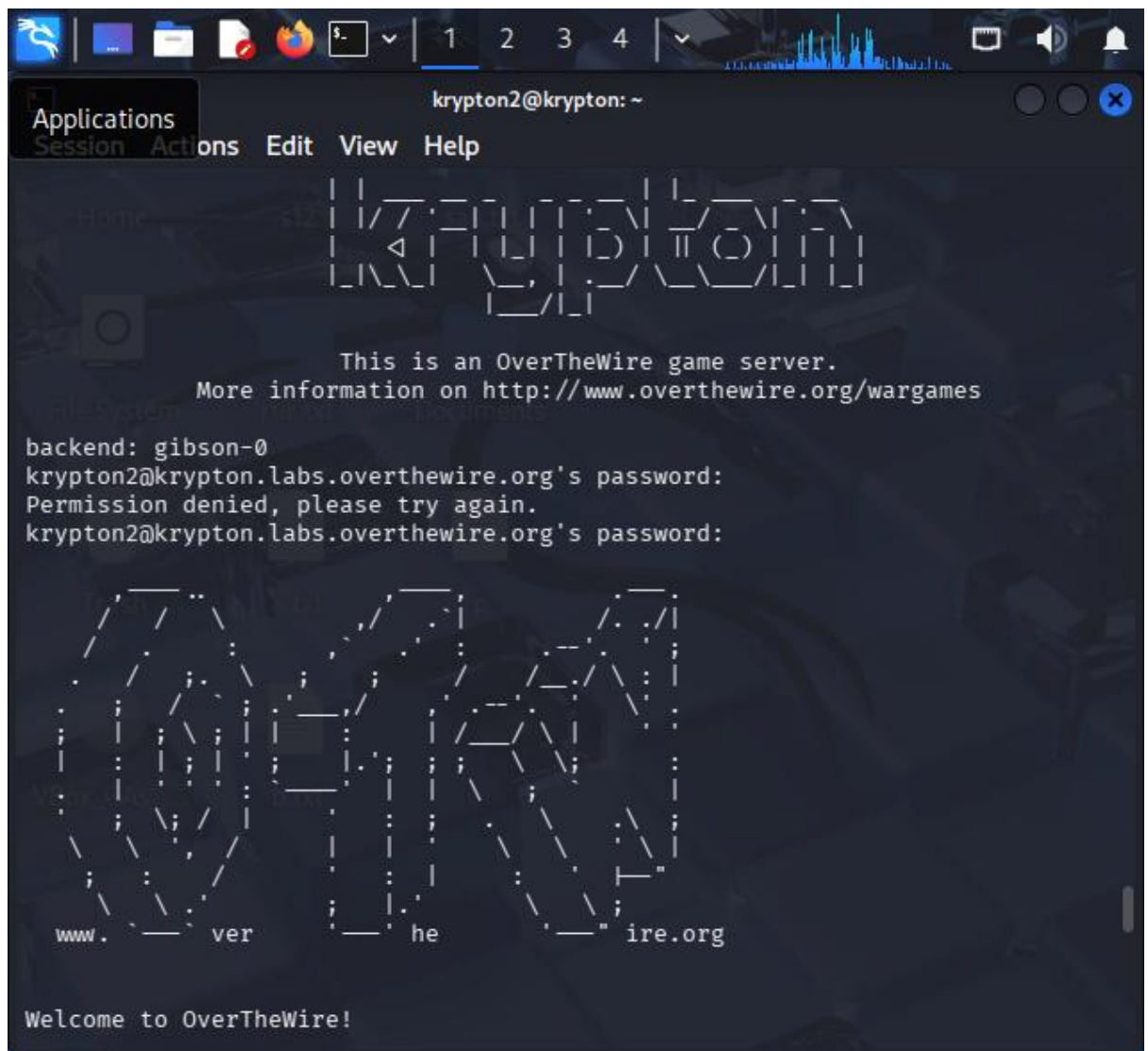
Krypton Level 2 → Level 3

Command: ssh,ls -la,cd,cat,mktemp,ls,touch,nano,tr

Steps:

9. Logged into the Krypton server using the credentials of level 2.
10. Listed the files in the home directory to check available files.
11. Navigated to the root directory to locate the Krypton folders.
12. Entered the Krypton directory and viewed all available level folders.
13. Moved into the krypton2 directory.
14. Read the README file to understand the level instructions and hints.
15. Displayed the encrypted file containing the password for the next level.
16. Created a temporary working directory in the /tmp location as suggested.
17. Switched into the newly created temporary directory.
18. Created a symbolic link to the encryption key file so it could be accessed.
19. Verified that the key file link was successfully created.
20. Executed the encryption program using a known system file to understand how encryption works.
21. Viewed the generated encrypted output file.

22. Created a plaintext file containing the English alphabet in uppercase.
23. Verified the contents of the plaintext file.
24. Encrypted the plaintext alphabet file using the encryption program.
25. Viewed the encrypted alphabet output.
26. Compared the plaintext alphabet with the encrypted alphabet to identify the letter shift.
27. Determined that the encryption method used was a Caesar cipher.
28. Applied the identified shift to decrypt the encrypted password file.
29. Successfully obtained the password for the next level.
30. Logged into the next level using the decrypted password.



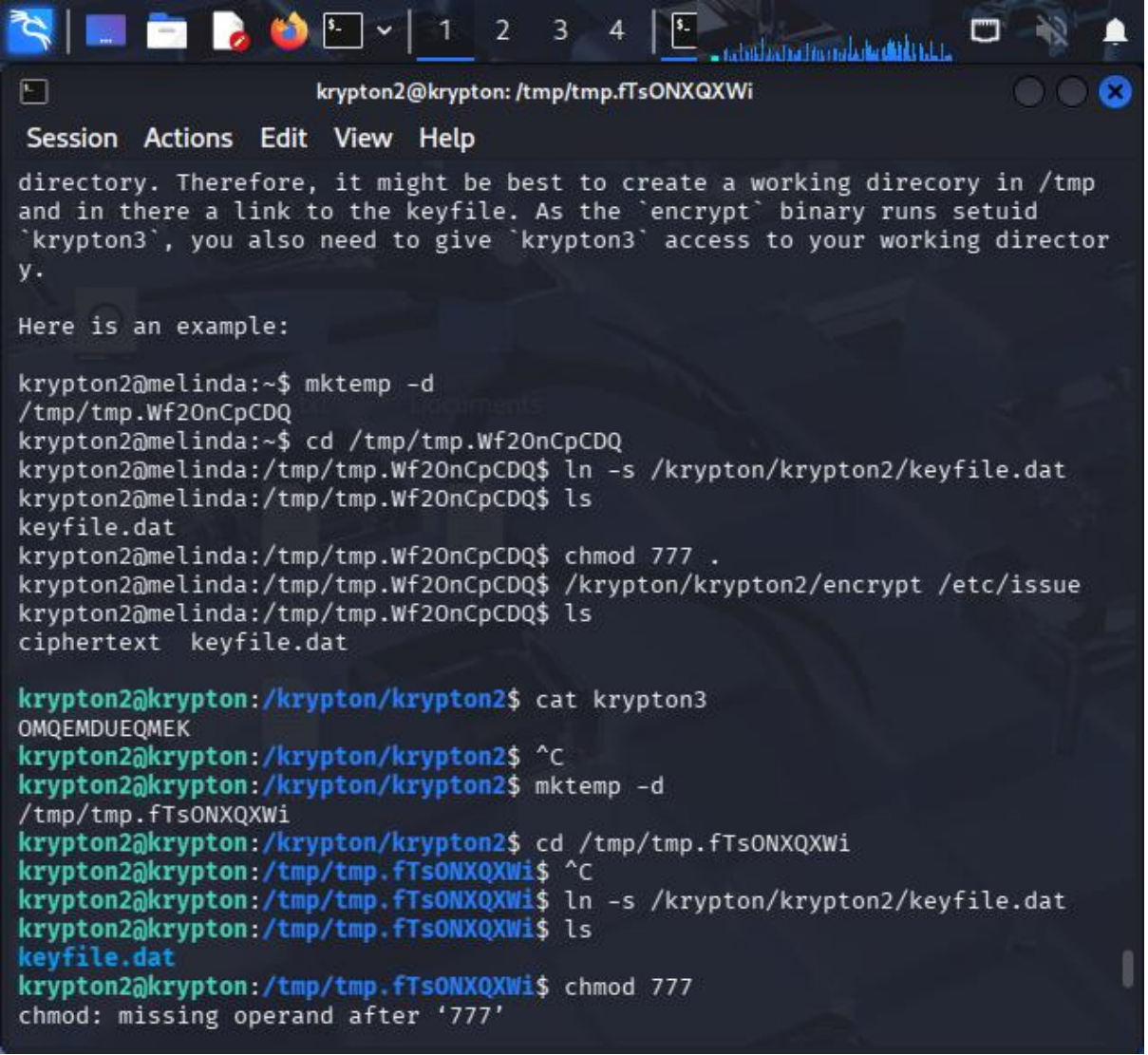
```
krypton2@krypton: /tmp/tmp.fTsONXQXWi
Session Actions Edit View Help

For support, questions or comments, contact us on discord or IRC.

Enjoy your stay!

krypton2@krypton:~$ ls
krypton2@krypton:~$
krypton2@krypton:~$ ls -la
total 20
drwxr-xr-x  2 root root 4096 Oct 14 09:27 .
drwxr-xr-x 150 root root 4096 Oct 14 09:29 ..
-rw-r--r--  1 root root  220 Mar 31 2024 .bash_logout
-rw-r--r--  1 root root 3851 Oct 14 09:19 .bashrc
-rw-r--r--  1 root root  807 Mar 31 2024 .profile
krypton2@krypton:~$ cd ../
krypton2@krypton:/$ ls
behemoth      krypton      media        snap
bin           lib          mnt          srv
bin.usr-is-merged lib32        narnia       sys
boot         lib64        opt          tmp
dev          lib.usr-is-merged proc         usr
drifter      libx32       root         utumno
etc          lost+found  run         var
formulaone   manpage     sbin        vortex
home        maze        sbin.usr-is-merged

krypton2@krypton:/$ cd /krypton/krypton2
krypton2@krypton:/krypton/krypton2$ ls
encrypt keyfile.dat krypton3 README
krypton2@krypton:/krypton/krypton2$ cat README
Krypton 2
```



A terminal window titled 'krypton2@krypton: /tmp/tmp.fTsONXQXWi' with standard window controls. The terminal shows a shell session where a directory is created, a symbolic link is made to a keyfile, permissions are set, and a file is encrypted. The user then switches to a different directory and repeats some steps, but encounters an error with the 'chmod' command.

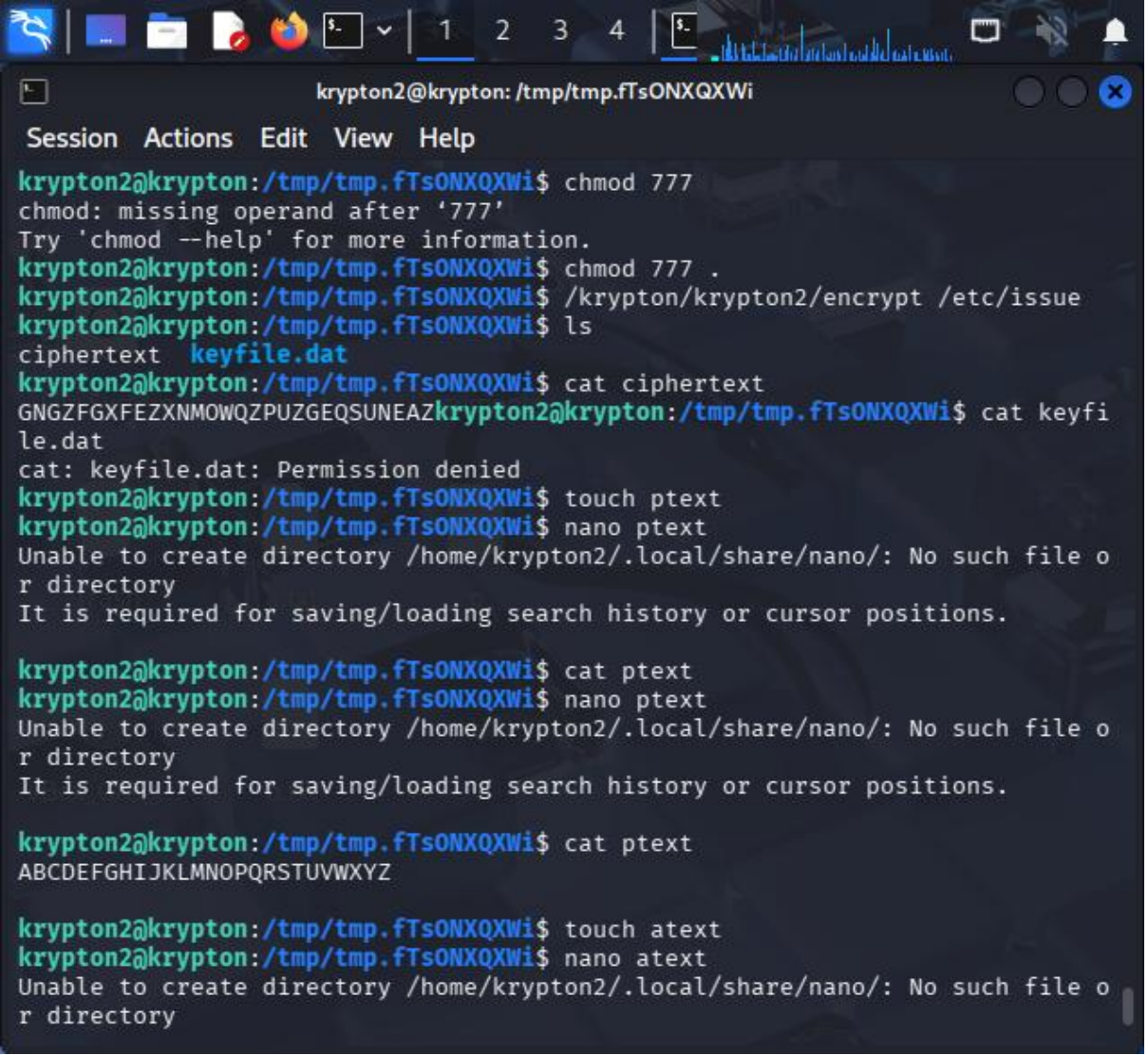
```
krypton2@krypton: /tmp/tmp.fTsONXQXWi
Session Actions Edit View Help

directory. Therefore, it might be best to create a working directory in /tmp
and in there a link to the keyfile. As the `encrypt` binary runs setuid
`krypton3`, you also need to give `krypton3` access to your working director
y.

Here is an example:

krypton2@melinda:~$ mkdir -p /tmp/tmp.Wf20nCpCDQ
krypton2@melinda:~$ cd /tmp/tmp.Wf20nCpCDQ
krypton2@melinda:/tmp/tmp.Wf20nCpCDQ$ ln -s /krypton/krypton2/keyfile.dat
krypton2@melinda:/tmp/tmp.Wf20nCpCDQ$ ls
keyfile.dat
krypton2@melinda:/tmp/tmp.Wf20nCpCDQ$ chmod 777 .
krypton2@melinda:/tmp/tmp.Wf20nCpCDQ$ /krypton/krypton2/encrypt /etc/issue
krypton2@melinda:/tmp/tmp.Wf20nCpCDQ$ ls
ciphertext keyfile.dat

krypton2@krypton:/krypton/krypton2$ cat krypton3
OMQEMDUEQMEK
krypton2@krypton:/krypton/krypton2$ ^C
krypton2@krypton:/krypton/krypton2$ mkdir -p
/tmp/tmp.fTsONXQXWi
krypton2@krypton:/krypton/krypton2$ cd /tmp/tmp.fTsONXQXWi
krypton2@krypton:/tmp/tmp.fTsONXQXWi$ ^C
krypton2@krypton:/tmp/tmp.fTsONXQXWi$ ln -s /krypton/krypton2/keyfile.dat
krypton2@krypton:/tmp/tmp.fTsONXQXWi$ ls
keyfile.dat
krypton2@krypton:/tmp/tmp.fTsONXQXWi$ chmod 777
chmod: missing operand after '777'
```

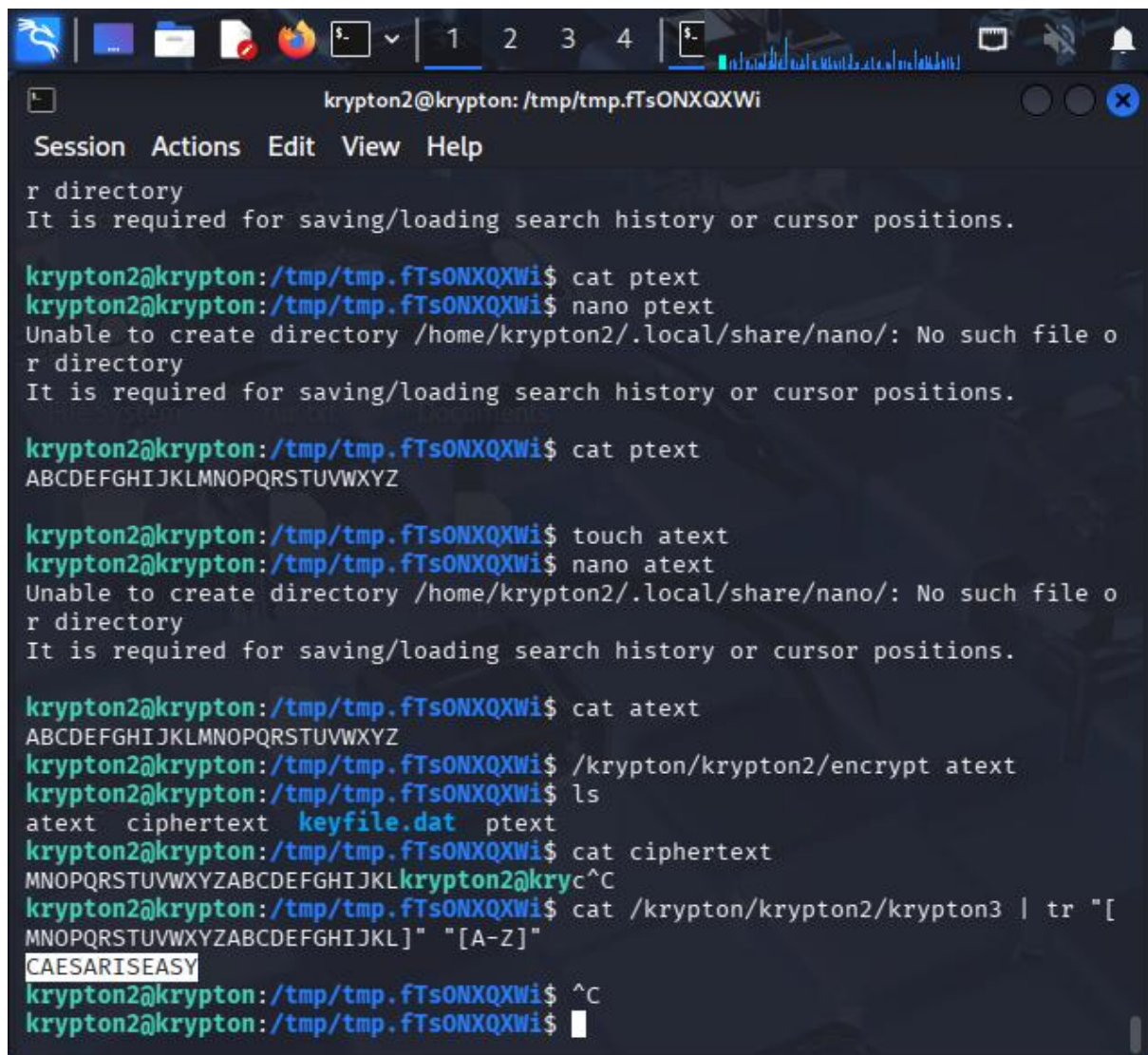
```
krypton2@krypton: /tmp/tmp.fTsONXQXWi
Session Actions Edit View Help

krypton2@krypton:/tmp/tmp.fTsONXQXWi$ chmod 777
chmod: missing operand after '777'
Try 'chmod --help' for more information.
krypton2@krypton:/tmp/tmp.fTsONXQXWi$ chmod 777 .
krypton2@krypton:/tmp/tmp.fTsONXQXWi$ /krypton/krypton2/encrypt /etc/issue
krypton2@krypton:/tmp/tmp.fTsONXQXWi$ ls
ciphertext  keyfile.dat
krypton2@krypton:/tmp/tmp.fTsONXQXWi$ cat ciphertext
GNGZFGXFEZXNMOWQZPUZGEQSUNEAZkrypton2@krypton:/tmp/tmp.fTsONXQXWi$ cat keyfi
le.dat
cat: keyfile.dat: Permission denied
krypton2@krypton:/tmp/tmp.fTsONXQXWi$ touch ptext
krypton2@krypton:/tmp/tmp.fTsONXQXWi$ nano ptext
Unable to create directory /home/krypton2/.local/share/nano/: No such file o
r directory
It is required for saving/loading search history or cursor positions.

krypton2@krypton:/tmp/tmp.fTsONXQXWi$ cat ptext
krypton2@krypton:/tmp/tmp.fTsONXQXWi$ nano ptext
Unable to create directory /home/krypton2/.local/share/nano/: No such file o
r directory
It is required for saving/loading search history or cursor positions.

krypton2@krypton:/tmp/tmp.fTsONXQXWi$ cat ptext
ABCDEFGHGIJKLMNOPQRSTUVWXYZ

krypton2@krypton:/tmp/tmp.fTsONXQXWi$ touch atext
krypton2@krypton:/tmp/tmp.fTsONXQXWi$ nano atext
Unable to create directory /home/krypton2/.local/share/nano/: No such file o
r directory
```



```
krypton2@krypton: /tmp/tmp.fTsONXQXWi
Session Actions Edit View Help

r directory
It is required for saving/loading search history or cursor positions.

krypton2@krypton: /tmp/tmp.fTsONXQXWi$ cat ptext
krypton2@krypton: /tmp/tmp.fTsONXQXWi$ nano ptext
Unable to create directory /home/krypton2/.local/share/nano/: No such file o
r directory
It is required for saving/loading search history or cursor positions.

krypton2@krypton: /tmp/tmp.fTsONXQXWi$ cat ptext
ABCDEFGHIJKLMNOPQRSTUVWXYZ

krypton2@krypton: /tmp/tmp.fTsONXQXWi$ touch atext
krypton2@krypton: /tmp/tmp.fTsONXQXWi$ nano atext
Unable to create directory /home/krypton2/.local/share/nano/: No such file o
r directory
It is required for saving/loading search history or cursor positions.

krypton2@krypton: /tmp/tmp.fTsONXQXWi$ cat atext
ABCDEFGHIJKLMNOPQRSTUVWXYZ
krypton2@krypton: /tmp/tmp.fTsONXQXWi$ /krypton/krypton2/encrypt atext
krypton2@krypton: /tmp/tmp.fTsONXQXWi$ ls
atext ciphertext keyfile.dat ptext
krypton2@krypton: /tmp/tmp.fTsONXQXWi$ cat ciphertext
MNOPQRSTUVWXYZABCDEFGHIJKLkrypton2@kryc^C
krypton2@krypton: /tmp/tmp.fTsONXQXWi$ cat /krypton/krypton2/krypton3 | tr "[
MNOPQRSTUVWXYZABCDEFGHIJKL]" "[A-Z]"
CAESARISEASY
krypton2@krypton: /tmp/tmp.fTsONXQXWi$ ^C
krypton2@krypton: /tmp/tmp.fTsONXQXWi$
```


9. Identify most frequent letters and patterns
10. Guess common English words (THE, LEVEL, PASSWORD, IS)
11. Replace cipher letters using character translation
12. View partially decrypted text
13. Refine substitutions until readable plaintext appears
14. Identify the final decrypted sentence
15. Extract the password for the next level
16. Logout from current level
17. Login to the next Krypton level using the discovered password

Code: import sys

```
if __name__=="__main__":
```

```
    char_table = {}
```

```
    char_total = 0
```

```
    groupsize = 0
```

```
    #Check Argument Counts
```

```
    if len(sys.argv) != 3:
```

```
        print("Usage: python3 freq_analysis.py filename groupsize")
```

```
        exit(0)
```

```
    else:
```

```
        try:
```

```
            groupsize = int(sys.argv[2])
```

```
        except:
```

```
            print("groupsize must be an int")
```

```
            exit(0)
```

```
    # Try to Open the specified file
```

```
    try:
```

```
        with open(sys.argv[1]) as fh:
```

```
            lines = fh.readlines()
```

```
    except:
```

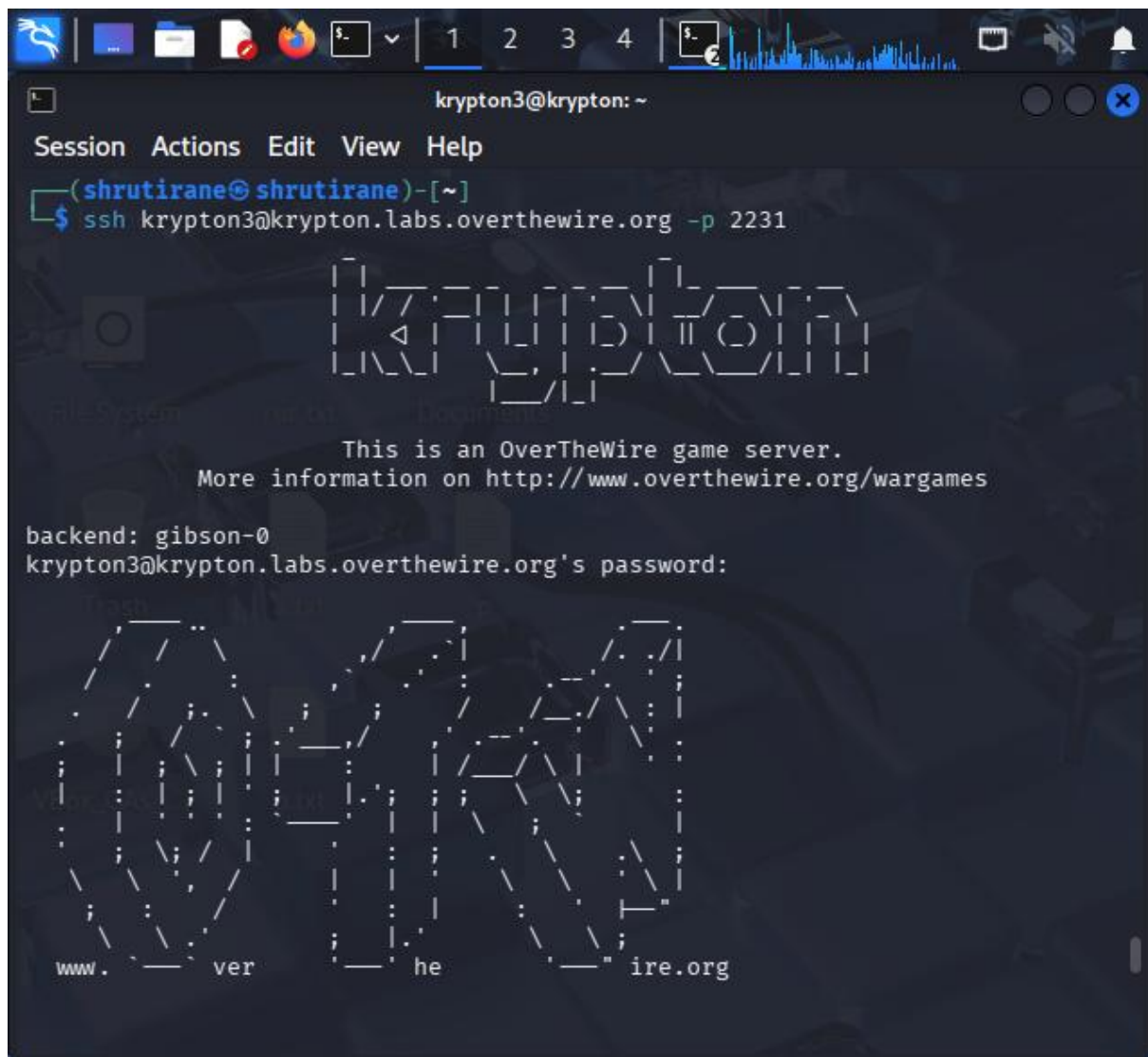
```
        print("No file named '" + sys.argv[1] + "'")
```

```
        exit(0)
```

```
for line in lines:
    line = line.replace(" ", "")
    line = line.replace("\n", "")
    for i in range(len(line) - groupsize):
        group = line[i:i+groupsize]
        if group in char_table:
            char_table[group] += 1
        else:
            char_table[group] = 1

char_table = sorted(char_table.items(), key=lambda x: x[1],
reverse=True)

for char in char_table:
    print(char[0] + ":\t" + str(char[1]))
```



A terminal window titled "krypton3@krypton: ~" with a menu bar (Session, Actions, Edit, View, Help) and window controls. The user "shrutirane" is at the prompt. The command executed is "ssh krypton3@krypton.labs.overthewire.org -p 2231". The output shows a large ASCII art logo for "OverTheWire", followed by the text "This is an OverTheWire game server. More information on <http://www.overthewire.org/wargames>". Below this, it says "backend: gibson-0" and "krypton3@krypton.labs.overthewire.org's password:". At the bottom, another large ASCII art logo for "OverTheWire" is displayed, with the URL "www.OverTheWire.org" at the very bottom.

```
krypton3@krypton: ~
Session Actions Edit View Help
(shrutirane@shrutirane)-[~]
$ ssh krypton3@krypton.labs.overthewire.org -p 2231

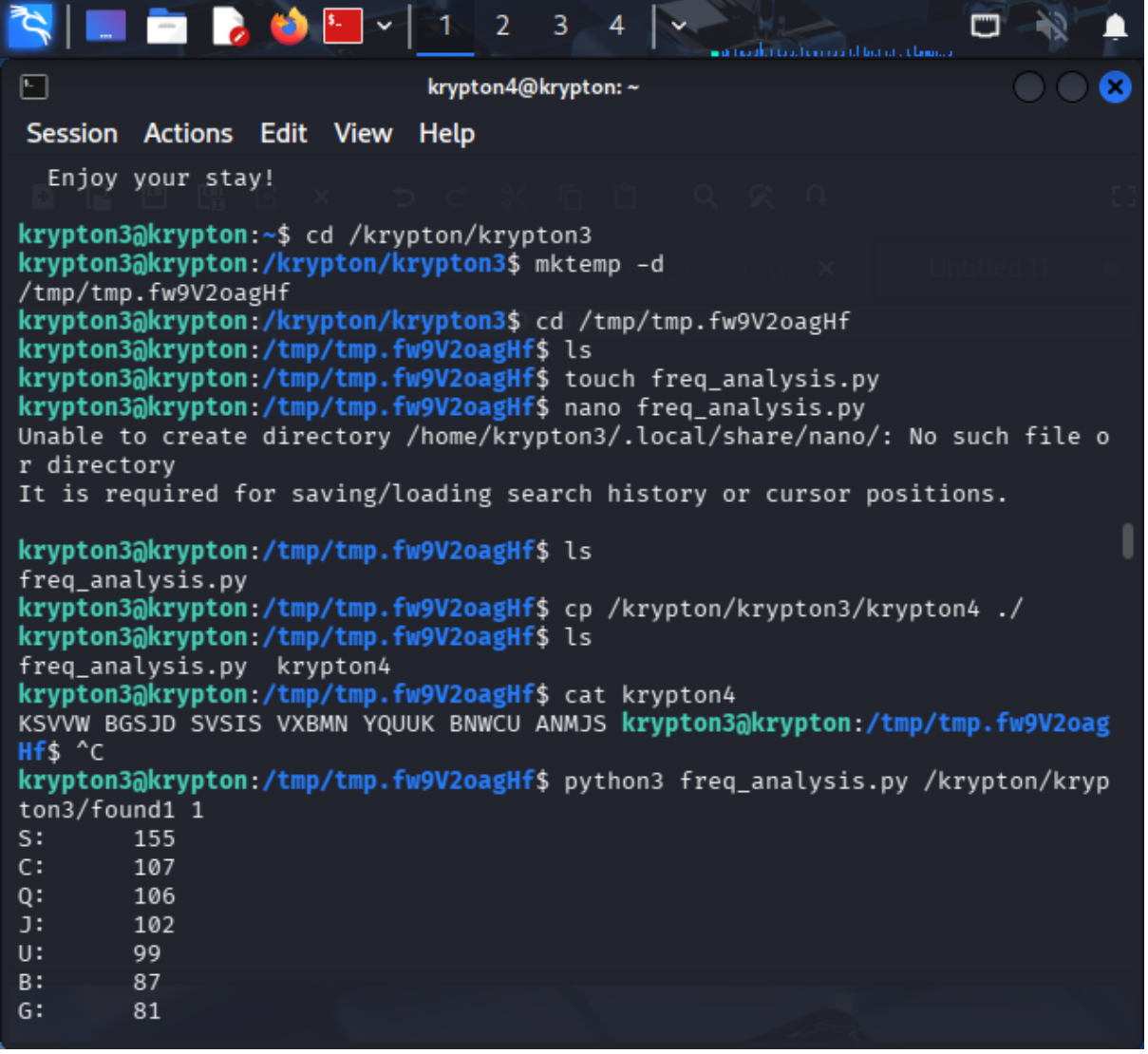
      _ _ _ _ _
     / /   / /
    / /   / /
   / /   / /
  / /   / /
 / /   / /
/_/_/_/_/_

This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames

backend: gibson-0
krypton3@krypton.labs.overthewire.org's password:

      _ _ _ _ _
     / /   / /
    / /   / /
   / /   / /
  / /   / /
 / /   / /
/_/_/_/_/_

www.OverTheWire.org
```



```
krypton4@krypton: ~
Session Actions Edit View Help

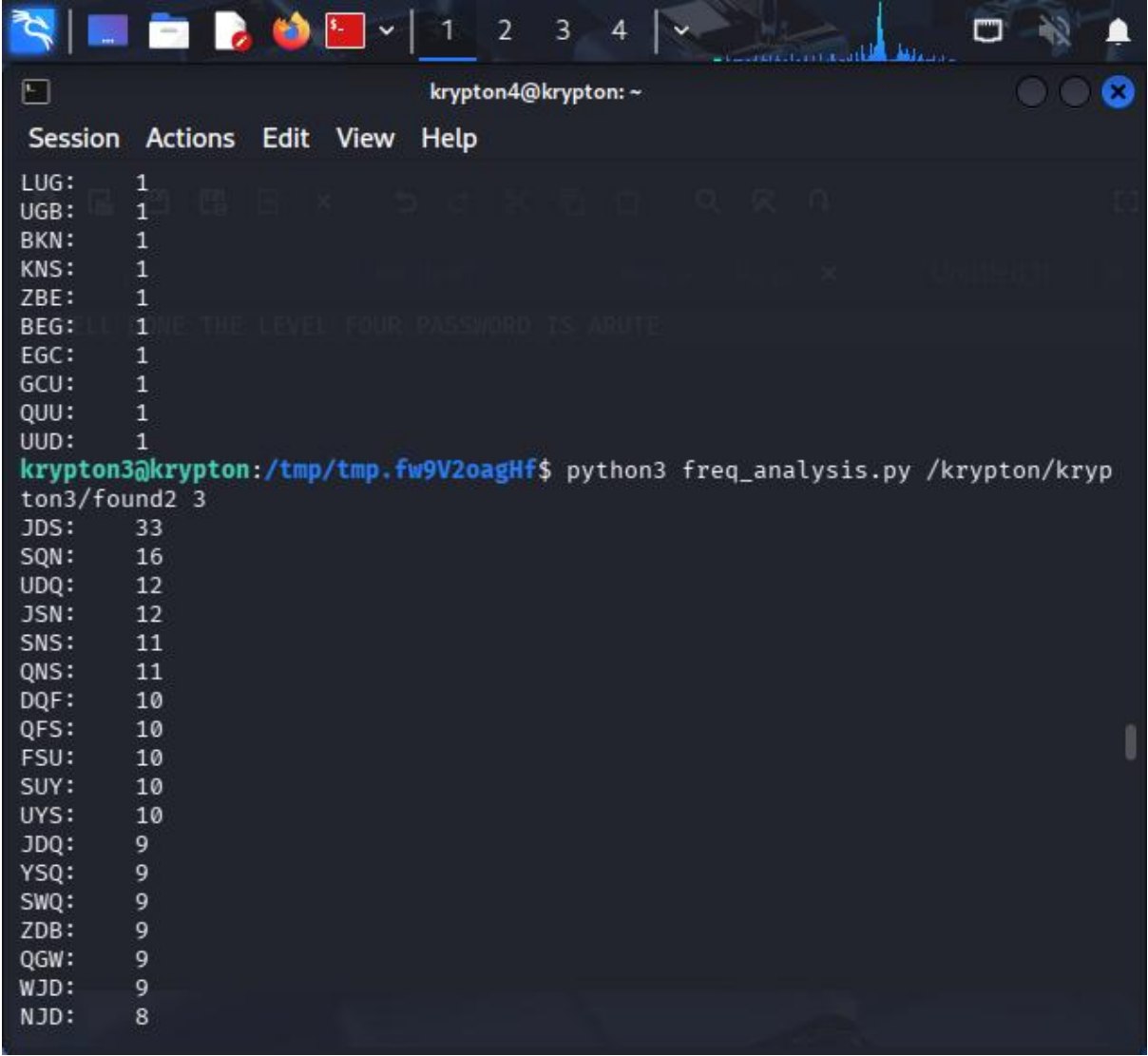
Enjoy your stay!

krypton3@krypton:~$ cd /krypton/krypton3
krypton3@krypton:/krypton/krypton3$ mktemp -d
/tmp/tmp.fw9V2oagHf
krypton3@krypton:/krypton/krypton3$ cd /tmp/tmp.fw9V2oagHf
krypton3@krypton:/tmp/tmp.fw9V2oagHf$ ls
krypton3@krypton:/tmp/tmp.fw9V2oagHf$ touch freq_analysis.py
krypton3@krypton:/tmp/tmp.fw9V2oagHf$ nano freq_analysis.py
Unable to create directory /home/krypton3/.local/share/nano/: No such file or directory
It is required for saving/loading search history or cursor positions.

krypton3@krypton:/tmp/tmp.fw9V2oagHf$ ls
freq_analysis.py
krypton3@krypton:/tmp/tmp.fw9V2oagHf$ cp /krypton/krypton3/krypton4 ./
krypton3@krypton:/tmp/tmp.fw9V2oagHf$ ls
freq_analysis.py  krypton4
krypton3@krypton:/tmp/tmp.fw9V2oagHf$ cat krypton4
KSVVW BGSJD SVSIS VXBMN YQUUK BNWCU ANMJS
krypton3@krypton:/tmp/tmp.fw9V2oagHf$ ^C
krypton3@krypton:/tmp/tmp.fw9V2oagHf$ python3 freq_analysis.py /krypton/krypton3/found1 1
S:      155
C:      107
Q:      106
J:      102
U:       99
B:       87
G:       81
```



```
krypton4@krypton: ~  
Session Actions Edit View Help  
krypton3@krypton:/tmp/tmp.fw9V2oagHf$ python3 freq_analysis.py /krypton/kryp  
ton3/found3 1  
S: 58  
Q: 48  
J: 40  
G: 35  
C: 34  
N: 31  
B: 30  
U: 27  
D: 22  
V: 21  
W: 16  
Z: 16  
E: 13  
M: 12  
K: 12  
Y: 9  
A: 9  
X: 9  
L: 6  
T: 6  
F: 5  
I: 3  
O: 2  
P: 1  
R: 1  
krypton3@krypton:/tmp/tmp.fw9V2oagHf$ python3 freq_analysis.py /krypton/kryp  
ton3/found1 3  
JDS: 19
```



```
krypton4@krypton: ~
Session Actions Edit View Help
LUG: 1
UGB: 1
BKN: 1
KNS: 1
ZBE: 1
BEG: 1 ONE THE LEVEL FOUR PASSWORD IS ARUTE
EGC: 1
GCU: 1
QUU: 1
UUD: 1
krypton3@krypton:/tmp/tmp.fw9V2oagHf$ python3 freq_analysis.py /krypton/kryp
ton3/found2 3
JDS: 33
SQN: 16
UDQ: 12
JSN: 12
SNS: 11
QNS: 11
DQF: 10
QFS: 10
FSU: 10
SUY: 10
UYS: 10
JDQ: 9
YSQ: 9
SWQ: 9
ZDB: 9
QGW: 9
WJD: 9
NJD: 8
```

```
WSQ: 1
krypton3@krypton:/tmp/tmp.fw9V2oagHf$ cat krypton4 | tr "[JDS]" "[THE]"
KEVVW BGETH EVEIE VXBMN YQUUK BNWCU ANMTE krypton3@krypton:/tmp/tmp.fw9V2oag
Hf$ ^C
krypton3@krypton:/tmp/tmp.fw9V2oagHf$ python3 freq_analysis.py /krypton/kryp
ton3/found2 1 THE LEVEL FOUR PASSWORD IS ARUTE
S: 243
Q: 186
J: 158
N: 135
U: 130
B: 129
D: 118
G: 111
C: 86
W: 66
Z: 59
V: 53
M: 45
T: 37
E: 34
X: 33
Y: 33
K: 30
L: 27
A: 26
I: 14
F: 12
O: 3
H: 2
```



```
krypton4@krypton: ~  
Session Actions Edit View Help  
krypton4@krypton.labs.overthewire.org's password:  
Connection closed by 51.21.210.216 port 2231  
  
(shrutirane@shrutirane)-[~]  
$ ssh krypton4@krypton.labs.overthewire.org -p 2231  
I WELL DONE THE LEVEL FOUR PASSWORD IS ABOVE  
  
KRYPTON  
  
This is an OverTheWire game server.  
More information on http://www.overthewire.org/wargames  
  
backend: gibson-0  
krypton4@krypton.labs.overthewire.org's password:
```

Krypton Level 3 → Level 4

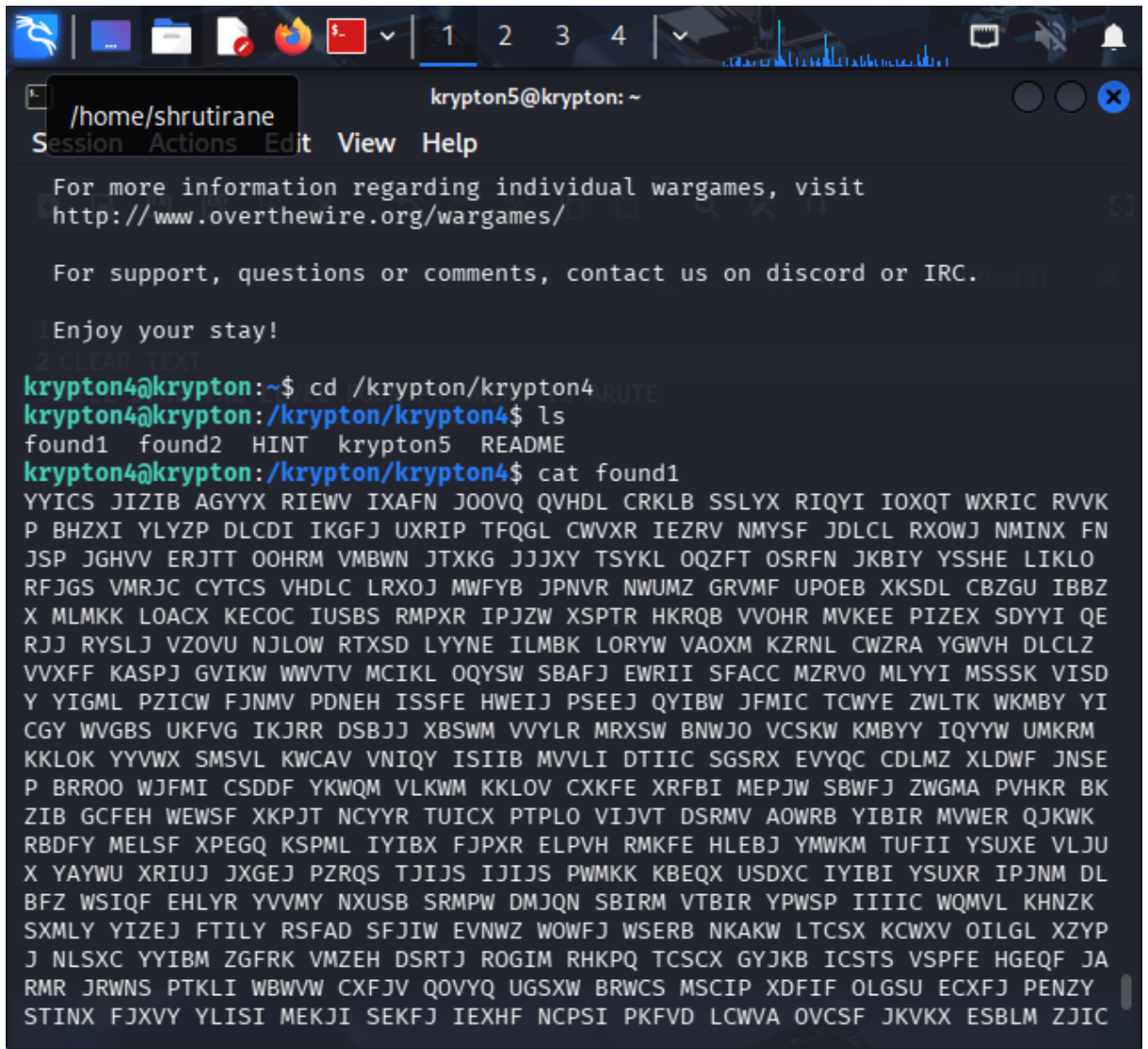
Command: cd, ls, cat, nano, ssh

Steps:

1. Connect to Krypton server using SSH
2. Open the terminal and navigate to the **krypton4** directory.
3. List the files and identify the encrypted file (**found1**).
4. Open the encrypted file and observe that it contains ciphertext.
5. Open an online Vigenère cipher decoder
6. Paste the ciphertext into the Vigenère decoder input field.
7. Use the given key **FREKEY** to decrypt the text.
8. Obtain the decrypted plaintext, which reveals the **password for krypton5**.

9. Return to the terminal and attempt to log in to **krypton5** using SSH.

10. Paste the decrypted password



```
/home/shrutirane krypton5@krypton: ~
Session Actions Edit View Help

For more information regarding individual wargames, visit
http://www.overthewire.org/wargames/

For support, questions or comments, contact us on discord or IRC.

Enjoy your stay!
1 CLEAR TEXT
krypton4@krypton:~$ cd /krypton/krypton4
krypton4@krypton:/krypton/krypton4$ ls
found1 found2 HINT krypton5 README
krypton4@krypton:/krypton/krypton4$ cat found1
YYICS JIZIB AGYYX RIEWV IXAFN JOOVQ QVHDL CRKLB SSLYX RIQYI IOXQT WXRIC RVVK
P BHZXI YLYZP DLCDI IKGFJ UXRIP TFQGL CWVXR IEZRV NMYSF JDLCL RXOWJ NMINX FN
JSP JGHVV ERJTT OOHBM VMBWN JTXKG JJJXY TSYKL OQZFT OSRFN JKBIY YSSHE LIKLO
RFJGS VMRJC CYTCS VHDLC LRXOJ MWFYB JPNVR NWUMZ GRVMF UPOEB XKSDL CBZGU IBBZ
X MLMKK LOACX KECOC IUSBS RMPXR IPJZW XSPTR HKRQB VVOHR MVKEE PIZEX SDYYI QE
RJJ RYSLJ VZOVU NJLOW RTXSD LYYNE ILMBK LORYW VAOXM KZRNL CWZRA YGWHV DLCLZ
VVXFF KASPJ GVIKW WWVTV MCIKL OQYSW SBAFJ EWRII SFACC MZRVO MLYYI MSSSK VISD
Y YIGML PZICW FJNMV PDNEH ISSFE HWEIJ PSEEJ QYIBW JFMIC TCWYE ZWLTK WKMBY YI
CGY WVGBS UKFVG IKJRR DSBJJ XBSWM VVYLR MRXSW BNWJO VCSKW KMBYY IQYYW UMKRM
KKLOK YYVWX SMSVL KWCAV VNIQY ISIIB MVVLI DTIIC SGSRX EVYQC CDLMZ XLDWF JNSE
P BRROO WJFMI CSDDF YKWQM VLKWM KKLOV CXKFE XRFBI MEPJW SBWFJ ZWGMA PVHKB BK
ZIB GCFEH WEWSF XKPJT NCYYR TUICX PTPLO VIJVT DSRMV AOWRB YIBIR MVWER QJKWK
RBDYF MELSF XPEGQ KSPML IYIBX FJPXR ELPVH RMKFE HLEBJ YMWKM TUFII YSUXE VLJU
X YAYWU XRIUJ JXGEJ PZRQS TJIJS IJIJS PWMKK KBEQX USDXC IYIBI YSUXR IPJNM DL
BFZ WSIQF EHLR YVVMY NXUSB SRMPW DMJQN SBIRM VTBIR YPWSP IIIIC WQMV LKHNZK
SXMLY YIZEJ FTILY RSFAD SFJIW EVNWZ WOWFJ WSERB NKAKW LTCSX KCWXV OILGL XZYP
J NLSXC YYIBM ZGFRK VMZEH DSRTJ ROGIM RHKPQ TCSCX GYJKB ICSTS VSPFE HGEQF JA
RMR JRWNS PTKLI WBWVW CXFJV QOVYQ UGSXW BRWCS MSCIP XDFIF OLGSU ECXFJ PENZY
STINX FJXVY YLISI MEKJI SEKFJ IEXHF NCPSI PKFVD LCWVA OVCSF JKVKX ESBLM ZJIC
```


Search for a tool

★ 🔍 SEARCH A TOOL ON dCODE

e.g. type 'random'

★ BROWSE THE FULL dCODE TOOLS' LIST

Results

📄 📋 📄 📄 📄 📄 📄 📄

🔑 FREKEY

ABCDEFGHIJKLMNOPQRSTUVWXYZ (26)

CLEAR TEXT

Vigenere Cipher - dCode

Tag(s) : Poly-Alphabetic Cipher

VIGENERE CIPHER

Cryptography › Poly-Alphabetic Cipher › Vigenere Cipher

VIGENERE DECODER



★ VIGENERE CIPHERTEXT

HCIKV RJOX

★ PLAINTEXT LANGUAGE English

★ ALPHABET ABCDEFGHIJKLMNOPQRSTUVWXYZ

► AUTOMATIC DECRYPTION

DECRYPTION METHOD

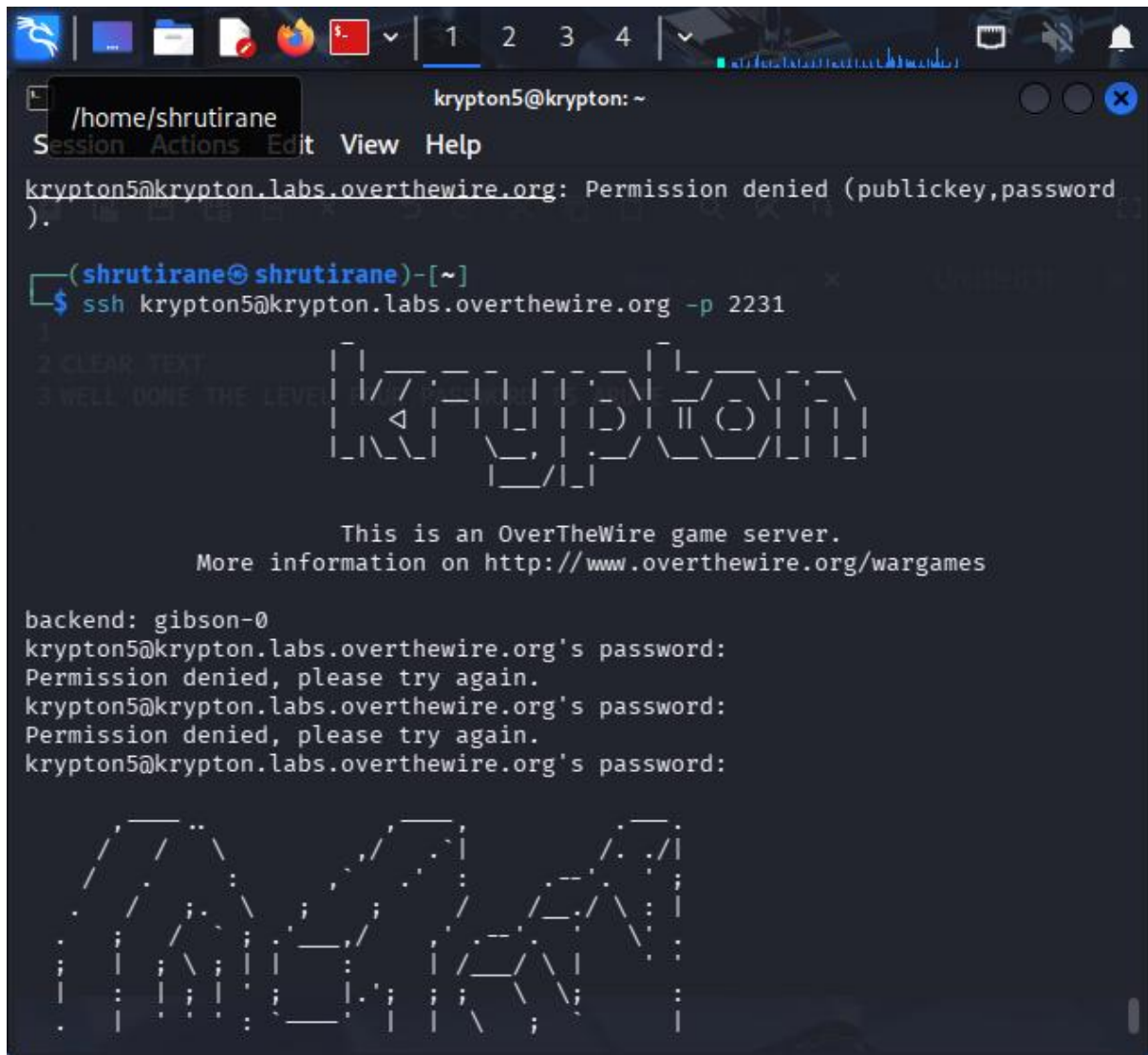
- ☒ KNOWING THE KEY/PASSWORD: FREKEY
- ☐ KNOWING THE KEY-LENGTH/SIZE, NUMBER OF LETTERS: 6
- ☐ KNOWING ONLY A PARTIAL KEY (JOKER=?): KE?
- ☐ KNOWING A PLAINTEXT WORD: CODE
- ☐ VIGENERE CRYPTANALYSIS (KASISKI'S TEST)

★ SHOW VIGENÈRE'S SQUARE/GRID (TABULA RECTA) ☐

► DECRYPT

See also: Autoclave Cipher – Beaufort Cipher – Caesar Cipher

VIGENERE ENCODER



```
/home/shrutirane krypton5@krypton: ~
Session Actions Edit View Help
krypton5@krypton.labs.overthewire.org: Permission denied (publickey,password).

(shrutirane@shrutirane)-[~]
$ ssh krypton5@krypton.labs.overthewire.org -p 2231
1
2 CLEAR TEXT
3 WELL DONE THE LEVEL
KRYPTON

This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames

backend: gibson-0
krypton5@krypton.labs.overthewire.org's password:
Permission denied, please try again.
krypton5@krypton.labs.overthewire.org's password:
Permission denied, please try again.
krypton5@krypton.labs.overthewire.org's password:
```

Krypton Level 4 → Level 5

Command:

ls,ls -la,cd,cat

Steps:

1. Open the terminal and listed files in the home directory.
2. Navigated up to the root directory.
3. Located and entered the krypton directory.
4. Entered the krypton5 directory.
5. Listed the contents of krypton5 to see available files.
6. Read the README file to understand the challenge description.

7. Identified that the cipher is a polyalphabetic (Vigenère) cipher with an unknown key length.
8. Displayed the encrypted text
9. Opened the Vigenère Cipher Decoder tool in a web browser.
10. Pasted the ciphertext into the Vigenère ciphertext input field.
11. Selected English as the plaintext language and kept the default alphabet.
12. Attempted decryption using frequency analysis / key-length guessing options.
13. Obtain the decrypted plaintext, which reveals the password for krypton6
14. Return to the terminal and attempt to log in to krypton6 using SSH.
15. Paste the decrypted password

```
1 2 3 4
/home/shrutirane krypton5@krypton: ~
Session Actions Edit View Help
krypton5@krypton.labs.overthewire.org: Permission denied (publickey,password).
(shrutirane@shrutirane)-[~]
$ ssh krypton5@krypton.labs.overthewire.org -p 2231
1
2 CLEAR TEXT
3 WELL DONE THE LEVEL
  KRYPTON
This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames
backend: gibson-0
krypton5@krypton.labs.overthewire.org's password:
Permission denied, please try again.
krypton5@krypton.labs.overthewire.org's password:
Permission denied, please try again.
krypton5@krypton.labs.overthewire.org's password:
  MAGE
```



```
krypton6@krypton: ~
Session Actions Edit View Help

krypton5@krypton:~$ ls
krypton5@krypton:~$ ls -la
total 20
drwxr-xr-x  2 root root 4096 Oct 14 09:27 .
drwxr-xr-x 150 root root 4096 Oct 14 09:29 ..
-rw-r--r--  1 root root  220 Mar 31 2024 .bash_logout
-rw-r--r--  1 root root 3851 Oct 14 09:19 .bashrc
-rw-r--r--  1 root root  807 Mar 31 2024 .profile
krypton5@krypton:~$ cd ../
krypton5@krypton:/$ ls
behemoth      krypton        media          snap
bin            lib            mnt            srv
bin.usr-is-merged lib32          narnia         sys
boot          lib64          opt            tmp
dev           lib.usr-is-merged proc           usr
drifter       libx32         root           utumno
etc           lost+found     run           var
formulaone    manpage       sbin          vortex
home          maze          sbin.usr-is-merged

krypton5@krypton:/$ cd /krypton
krypton5@krypton:/krypton$ ls
krypton1  krypton2  krypton3  krypton4  krypton5  krypton6  krypton7
krypton5@krypton:/krypton$ cd krypton5
krypton5@krypton:/krypton/krypton5$ ls
found1  found2  found3  krypton6  README
krypton5@krypton:/krypton/krypton5$ cat krypton6
BELOS Z
krypton5@krypton:/krypton/krypton5$ cat README
Frequency analysis can break a known key length as well. Lets try one
last polyalphabetic cipher, but this time the key length is unknown.
```



Search for a tool

★ 🔍 SEARCH A TOOL ON dCODE

e.g. type 'random'

★ BROWSE THE FULL dCODE TOOLS' LIST

Results

🔑 KEYLENGTH

ABCDEFGHIJKLMNOPQRSTUVWXYZ (26)

RANDOM

Vigenere Cipher - [dCode](#)

Tao(s) : Poly-Alphabetic Cipher

VIGENERE CIPHER

Cryptography › Poly-Alphabetic Cipher › Vigenere Cipher

VIGENERE DECRYPTOR

★ VIGENERE CIPHERTEXT

BELOS Z

★ PLAINTEXT LANGUAGE English

★ ALPHABET ABCDEFGHIJKLMNOPQRSTUVWXYZ

AUTOMATIC DECRYPTION

DECRYPTION METHOD

☒ KNOWING THE KEY/PASSWORD: KEYLENGTH

☐ KNOWING THE KEY-LENGTH/SIZE, NUMBER OF LETTERS: 6

☐ KNOWING ONLY A PARTIAL KEY (JOKER=?): KE?

☐ KNOWING A PLAINTEXT WORD: CODE

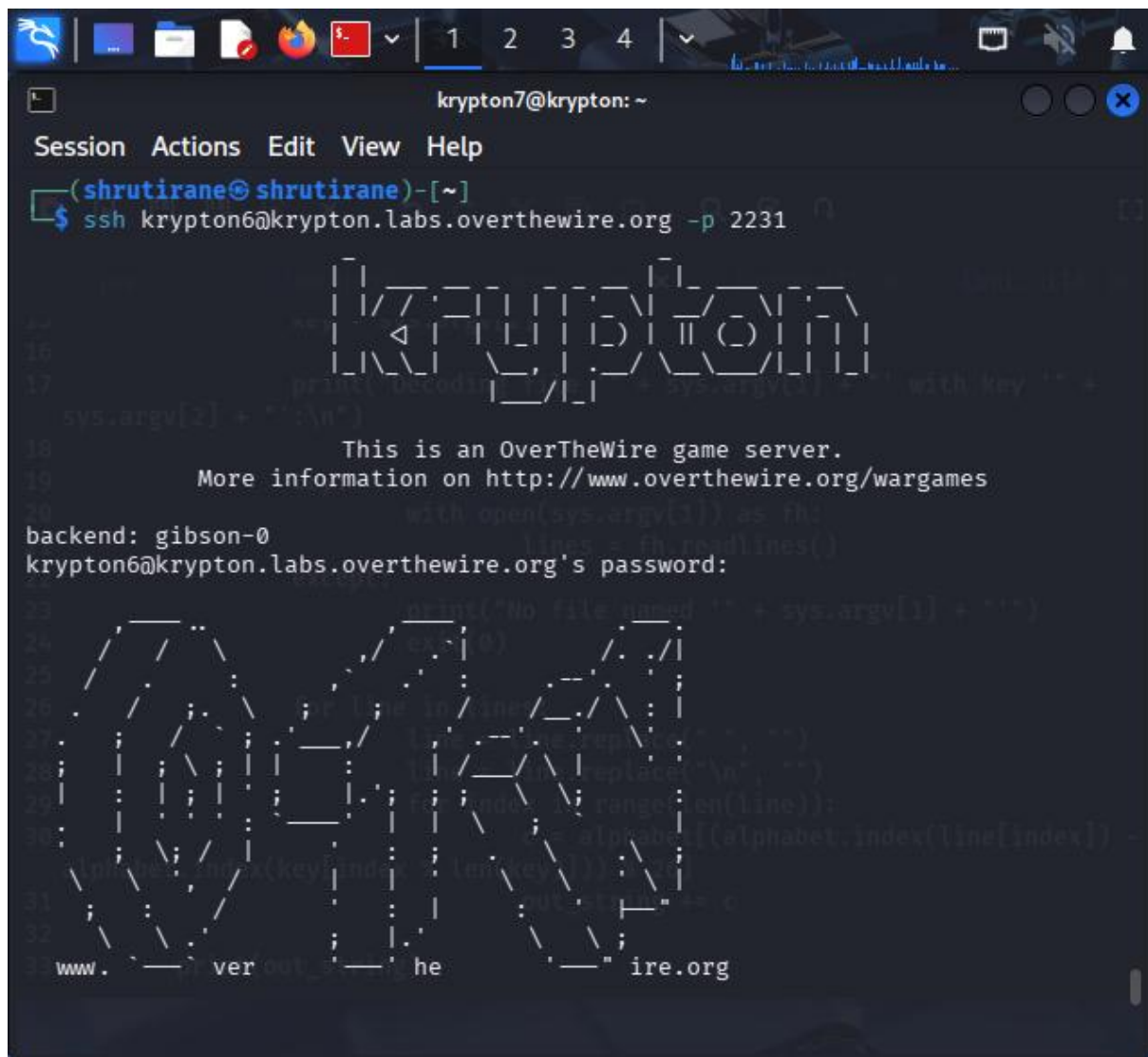
☐ VIGENERE CRYPTANALYSIS (KASISKI'S TEST)

★ SHOW VIGENÈRE'S SQUARE/GRID (TABULA RECTA)

DECRYPT

See also: Autoclave Cipher – Beaufort Cipher – Caesar Cipher

VIGENERE ENCRYPTOR



The screenshot shows a terminal window titled 'krypton7@krypton: ~'. The user 'shrutirane' is logged in. They have executed the command `ssh krypton6@krypton.labs.overthewire.org -p 2231`. The terminal output displays a large ASCII art logo for 'OverTheWire' and the following text: 'This is an OverTheWire game server. More information on <http://www.overthewire.org/wargames>'. Below this, it says 'backend: gibson-0' and 'krypton6@krypton.labs.overthewire.org's password:'. At the bottom, another ASCII art logo for 'OverTheWire' is shown, with the URL 'www.overthewire.org' at the very bottom.

Krypton Level 5 → Level 6

Command: touch ,nano , cat , ls ,xxd ,encrypt6 , python3 , exit

Steps :

1. Navigate to a temporary working directory.
2. Create a plain text file.
3. Open the text file and enter repeated characters (letter "A"), then save it.(a.txt)
4. Display the contents of the plain text file to verify it.
5. Encrypt the plain text file using the provided encryption program and a key file.

6. Confirm that a new encrypted (cipher) file is created.
7. Examine the binary representation of the original plain text file.
8. Examine the binary representation of the encrypted file.
9. Copy the encrypted output into a new file for decoding.
10. Create a decoder script file.(vignere_decoder.py)
11. List directory contents to confirm all required files exist.
12. Run the decoder using the encrypted file and the extracted key.
13. Obtain the decoded plaintext output.
14. Exit session.

Code: import sys

key = ""

alphabet = "ABCDEFGHIJKLMNOPQRSTUVWXYZ"

if __name__=="__main__":

 out_string = ""

 if (len(sys.argv)) != 3:

 print("Usage: python3 vignere_decoder.py filename key")

 exit(0)

 else:

 key = sys.argv[2]

 print("Decoding file " + sys.argv[1] + " with key " + sys.argv[2] + "\n")

 try:

 with open(sys.argv[1]) as fh:

 lines = fh.readlines()

 except:

 print("No file named " + sys.argv[1] + "")

 exit(0)

 for line in lines:

```
line = line.replace(" ", "")
        line = line.replace("\n", "")
        for index in range(len(line)):
            c = alphabet[(alphabet.index(line[index]) -
alphabet.index(key[index % len(key)])) % 26]
            out_string += c

        print(out_string)
```

```
krypton7@krypton: ~
Applications Session Actions Edit View Help

encrypt6 HINT1 HINT2 keyfile.dat krypton7 onetime README
krypton6@krypton:/tmp/tmp.pPJ37WSgcb$ touch a.txt
krypton6@krypton:/tmp/tmp.pPJ37WSgcb$ nano a.txt
Unable to create directory /home/krypton6/.local/share/nano/: No such file or directory
It is required for saving/loading search history or cursor positions.

krypton6@krypton:/tmp/tmp.pPJ37WSgcb$ nano a.txt
Unable to create directory /home/krypton6/.local/share/nano/: No such file or directory
It is required for saving/loading search history or cursor positions.

krypton6@krypton:/tmp/tmp.pPJ37WSgcb$ cat a.txt
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAa
krypton6@krypton:/tmp/tmp.pPJ37WSgcb$ /krypton/krypton6/encrypt6 a.txt ciphe
r.txt
krypton6@krypton:/tmp/tmp.pPJ37WSgcb$ ls
a.txt cipher.txt keyfile.dat
krypton6@krypton:/tmp/tmp.pPJ37WSgcb$ cat cipher.txt
EICTDGYIYZKTHNSIRFXYPFUEOCKRNEICTDGYIYZKTHNSIRFXYPFUEOCKRNEICTkrypton6@kry
pton:/tmp/tmp.pPJ37WSgcb$ cat ciphera
cat: ciphera: No such file or directory
krypton6@krypton:/tmp/tmp.pPJ37WSgcb$ xxd -b a.txt
00000000: 01000001 01000001 01000001 01000001 01000001 01000001 01000001 01000001  AAAAAA
00000006: 01000001 01000001 01000001 01000001 01000001 01000001 01000001 01000001  AAAAAA
0000000c: 01000001 01000001 01000001 01000001 01000001 01000001 01000001 01000001  AAAAAA
00000012: 01000001 01000001 01000001 01000001 01000001 01000001 01000001 01000001  AAAAAA
00000018: 01000001 01000001 01000001 01000001 01000001 01000001 01000001 01000001  AAAAAA
0000001e: 01000001 01000001 01000001 01000001 01000001 01000001 01000001 01000001  AAAAAA
00000024: 01000001 01000001 01000001 01000001 01000001 01000001 01000001 01000001  AAAAAA
```

```
krypton7@krypton: ~
Session Actions Edit View Help

It is required for saving/loading search history or cursor positions.

krypton6@krypton:/tmp/tmp.pPJ37WSgcb$ cat a.txt
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAa
krypton6@krypton:/tmp/tmp.pPJ37WSgcb$ /krypton/krypton6/encrypt6 a.txt cipher.txt
krypton6@krypton:/tmp/tmp.pPJ37WSgcb$ ls
a.txt cipher.txt keyfile.dat
krypton6@krypton:/tmp/tmp.pPJ37WSgcb$ cat cipher.txt
EICTDGYIYZKTHNSIRFXPCFUEOCKRNEICTDGYIYZKTHNSIRFXPCFUEOCKRNEICTkrypton6@kry
pton:/tmp/tmp.pPJ37WSgcb$ cat ciphera
cat: ciphera: No such file or directory
krypton6@krypton:/tmp/tmp.pPJ37WSgcb$ xxd -b a.txt
00000000: 01000001 01000001 01000001 01000001 01000001 01000001 01000001 01000001  AAAAAA
00000006: 01000001 01000001 01000001 01000001 01000001 01000001 01000001 01000001  AAAAAA
0000000c: 01000001 01000001 01000001 01000001 01000001 01000001 01000001 01000001  AAAAAA
00000012: 01000001 01000001 01000001 01000001 01000001 01000001 01000001 01000001  AAAAAA
00000018: 01000001 01000001 01000001 01000001 01000001 01000001 01000001 01000001  AAAAAA
0000001e: 01000001 01000001 01000001 01000001 01000001 01000001 01000001 01000001  AAAAAA
00000024: 01000001 01000001 01000001 01000001 01000001 01000001 01000001 01000001  AAAAAA
0000002a: 01000001 01000001 01000001 01000001 01000001 01000001 01000001 01000001  AAAAAA
00000030: 01000001 01000001 01000001 01000001 01000001 01000001 01000001 01000001  AAAAAA
00000036: 01000001 01000001 01000001 01000001 01000001 01000001 01000001 01000001  AAAAAA
0000003c: 01000001 01000001 01000001 01100001 00001010 00001010 00001010 00001010  AAAa.
krypton6@krypton:/tmp/tmp.pPJ37WSgcb$ xxd -b cipher.txt
00000000: 01000101 01001001 01000011 01010100 01000100 01000111 01000111 01000111  EICTDG
00000006: 01011001 01001001 01011001 01011010 01001011 01010100 01010100 01010100  YIYZKT
0000000c: 01001000 01001110 01010011 01001001 01010010 01000110 01000110 01000110  HNSIRF
00000012: 01011000 01011001 01000011 01010000 01000110 01010101 01010101 01010101  XYCPFU
00000018: 01000101 01001111 01000011 01001011 01010010 01010010 01001110 01001110  EOCKRN
```



```

krypton7@krypton: ~
Session  Actions  Edit  View  Help
0000001e: 01000101 01001001 01000011 01010100 01000100 01000111 EICTDG
00000024: 01011001 01001001 01011001 01011010 01001011 01010100 YIYZKT
0000002a: 01001000 01001110 01010011 01001001 01010010 01000110 HNSIRF
00000030: 01011000 01011001 01000011 01010000 01000110 01010101 XYCPFU
00000036: 01000101 01001111 01000011 01001011 01010010 01001110 EOCKRN
0000003c: 01000101 01001001 01000011 01010100 EICT
krypton6@krypton:/tmp/tmp.pPJ37WSgcb$ /krypton/krypton6/encrypt6 a.txt ciphe
r.txt_a.txt
krypton6@krypton:/tmp/tmp.pPJ37WSgcb$ cat a.txt
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
krypton6@krypton:/tmp/tmp.pPJ37WSgcb$ cat cipher.txt
EICTDGYIYZKTHNSIRFXPCFUEOCKRNEICTDGYIYZKTHNSIRFXPCFUEOCKRNEICTkrypton6@kry
pton:/tmp/tmp.pPJ37WSgcb$ cat cipher.txt_a.txt
EICTDGYIYZKTHNSIRFXPCFUEOCKRNEICTDGYIYZKTHNSIRFXPCFUEOCKRNEICTkrypton6@kry
pton:/tmp/tmp.pPJ37WSgcb$ touch vignere_decoder.py
krypton6@krypton:/tmp/tmp.pPJ37WSgcb$ nano vignere_decoder.py
Unable to create directory /home/krypton6/.local/share/nano/: No such file o
r directory
It is required for saving/loading search history or cursor positions.
krypton6@krypton:/tmp/tmp.pPJ37WSgcb$ ls
a.txt  cipher.txt  cipher.txt_a.txt  keyfile.dat  vignere_decoder.py
krypton6@krypton:/tmp/tmp.pPJ37WSgcb$ python3 vignere_decoder.py /krypton/kr
ypton6/krypton7 EICTDGYIYZKTHNSIRFXPCFUEOCKRN
Decoding file '/krypton/krypton6/krypton7' with key 'EICTDGYIYZKTHNSIRFXPCF
UEOCKRN':

LFSRISNOTRANDOM (out_string)
krypton6@krypton:/tmp/tmp.pPJ37WSgcb$ exit
logout

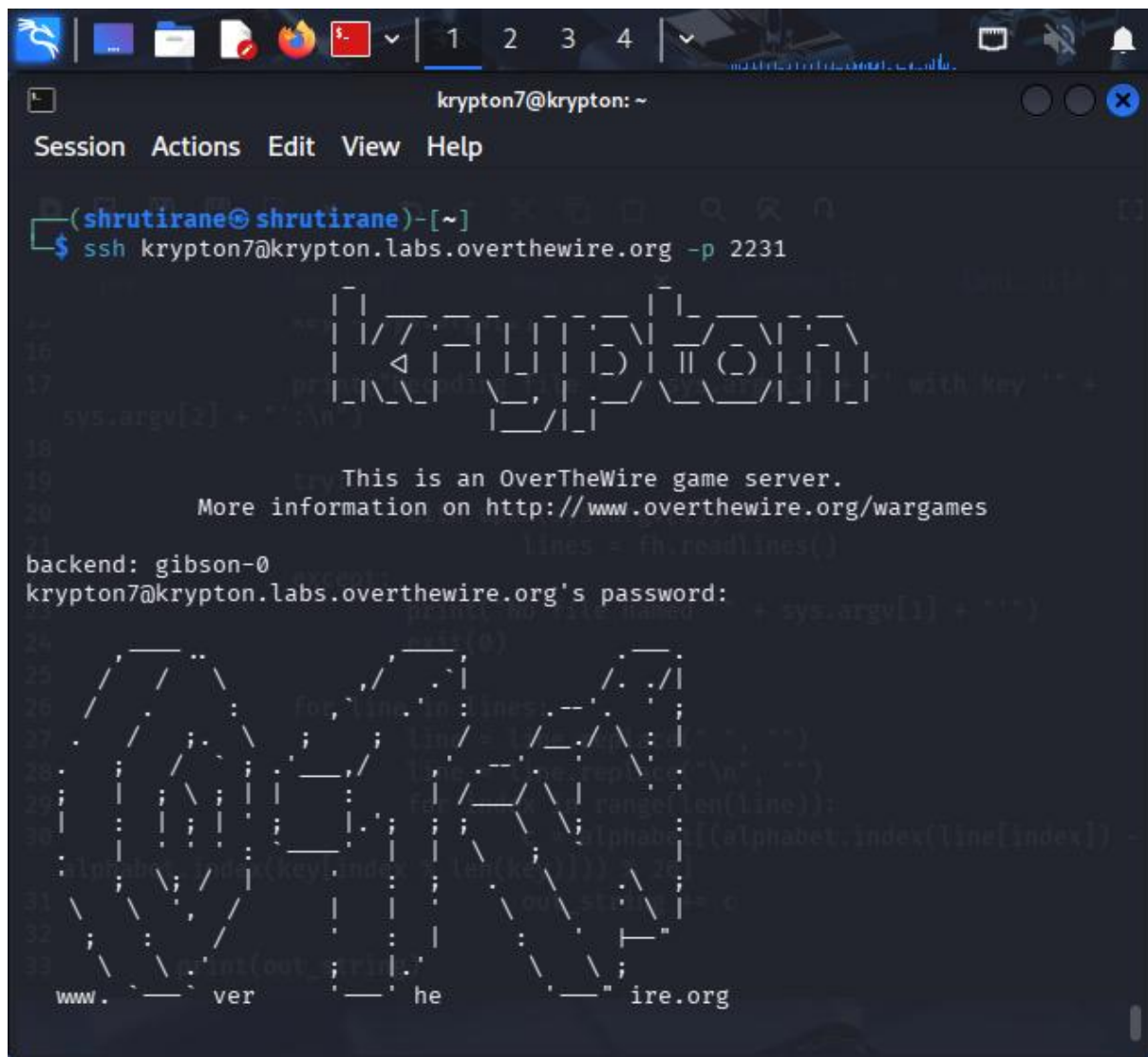
```

Krypton Level 6 → Level 7

Command: ls -la ,cd,cat

Steps:

- 1)login using password that got in previous .
- 2)List the folders .
- 3)go to main directory.
- 4)Open the krypton7 and README file ,to get message of finishing the level.



A terminal window titled "krypton7@krypton: ~" with a menu bar (Session, Actions, Edit, View, Help) and a toolbar. The terminal shows a user "shrutirane" at a prompt. They execute the command "ssh krypton7@krypton.labs.overthewire.org -p 2231". The connection is successful, and the user is greeted with ASCII art of the word "WELCOME" and a message: "This is an OverTheWire game server. More information on http://www.overthewire.org/wargames". The user is then prompted for a password. The terminal also shows some Python code in the background, including "sys.argv[2] + *", "lines = fh.readlines()", "print('Welcome to the game server. The password is: ' + sys.argv[1] + '*')", "for line in lines:", "line = line.strip()", "for i in range(len(line)):", "alphabet[(alphabet.index(line[i]) - alphabet.index(key[i])) % 26]", "out_string += c", and "print(out_string)". At the bottom, the URL "www.OverTheWire.org" is displayed in ASCII art.

```
krypton7@krypton: ~
Session Actions Edit View Help

(shrutirane@shrutirane)-[~]
$ ssh krypton7@krypton.labs.overthewire.org -p 2231

WELCOME

This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames

backend: gibson-0
krypton7@krypton.labs.overthewire.org's password:

www.OverTheWire.org
```

```
krypton7@krypton: /krypton/krypton7
Session Actions Edit View Help

krypton7@krypton:~$ ls -la
total 20
drwxr-xr-x  2 root root 4096 Oct 14 09:27 .
drwxr-xr-x 150 root root 4096 Oct 14 09:29 ..
-rw-r--r--  1 root root  220 Mar 31 2024 .bash_logout
-rw-r--r--  1 root root 3851 Oct 14 09:19 .bashrc
-rw-r--r--  1 root root  807 Mar 31 2024 .profile
krypton7@krypton:~$ cd ../
krypton7@krypton:/$ ls
behemoth      krypton        media          snap
bin            lib            mnt            srv
bin.usr-is-merged lib32          narnia         sys
boot          lib64          opt            tmp
dev           lib.usr-is-merged proc           usr
drifter       libx32         root           utumno
etc           lost+found    run           var
formulaone    manpage       sbin          vortex
home          maze          sbin.usr-is-merged

krypton7@krypton:/$ cd /krypton
krypton7@krypton:/krypton$ ls
krypton1  krypton2  krypton3  krypton4  krypton5  krypton6  krypton7
krypton7@krypton:/krypton$ cd /krypton7
-bash: cd: /krypton7: No such file or directory
krypton7@krypton:/krypton$ cd krypton7
krypton7@krypton:/krypton/krypton7$ ls
README
krypton7@krypton:/krypton/krypton7$ cat README
Congratulations on beating Krypton!
krypton7@krypton:/krypton/krypton7$
```